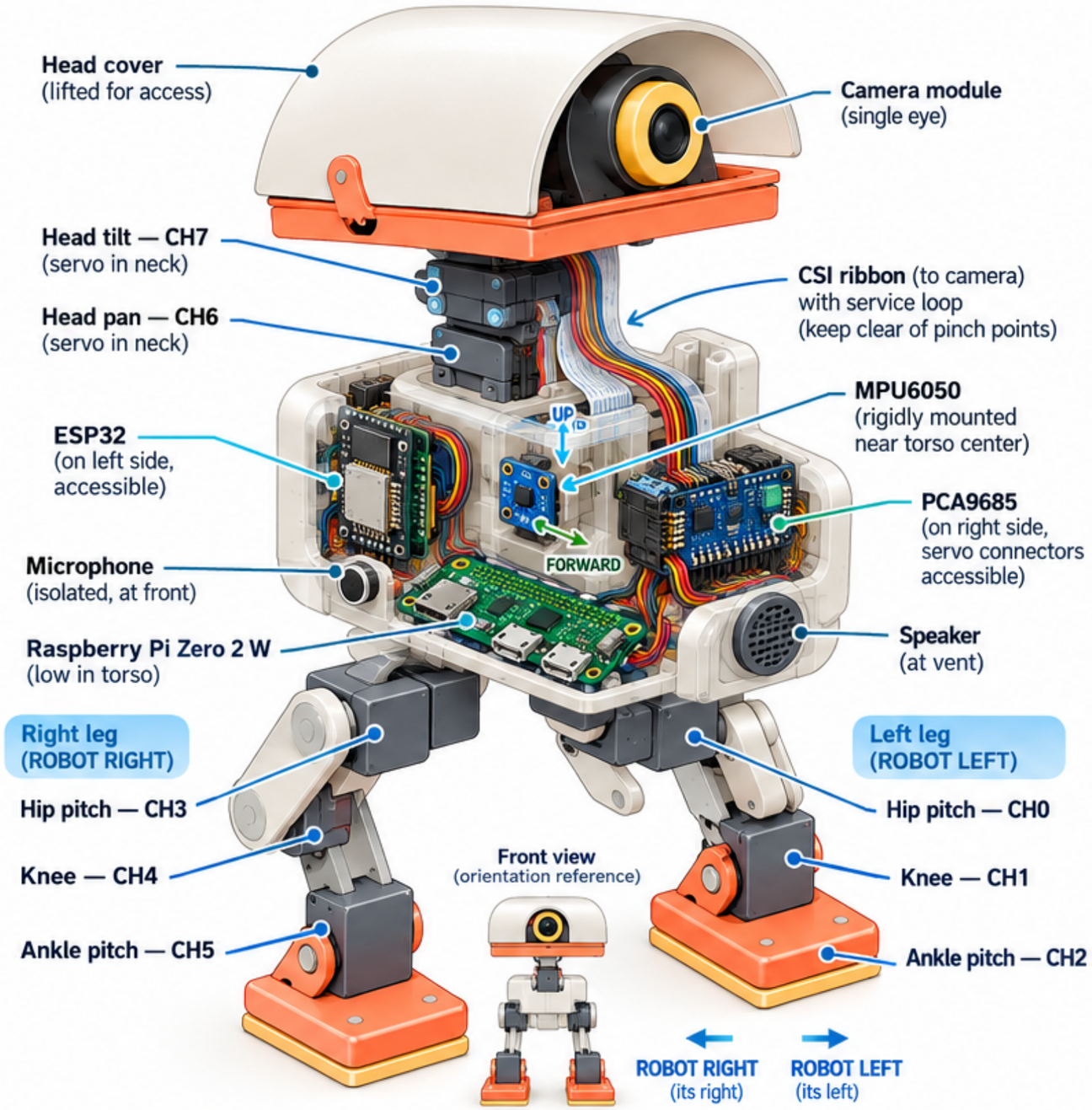
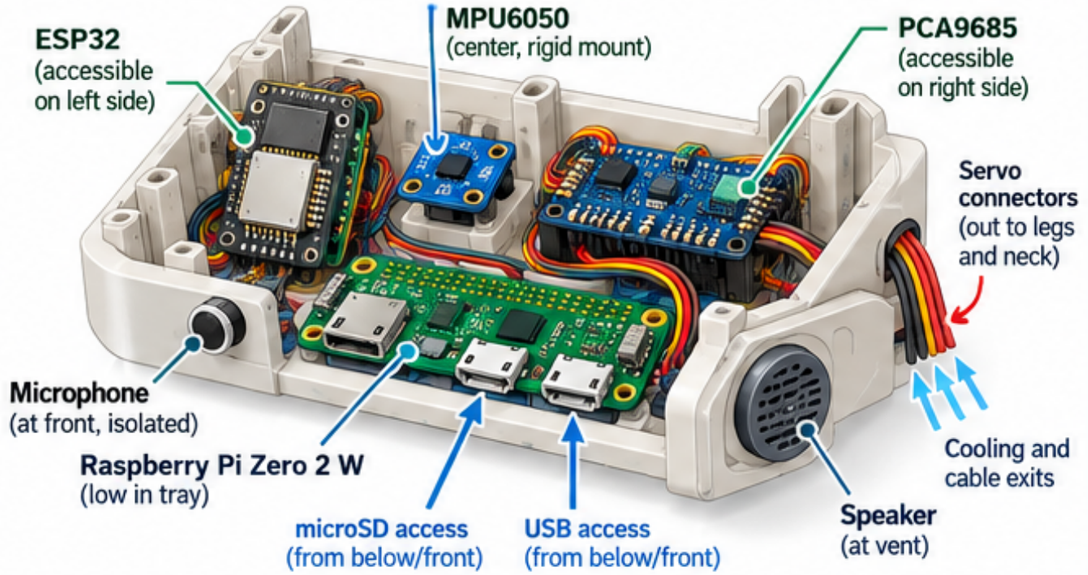
MICRODUCK DIY

WHERE EVERYTHING GOES

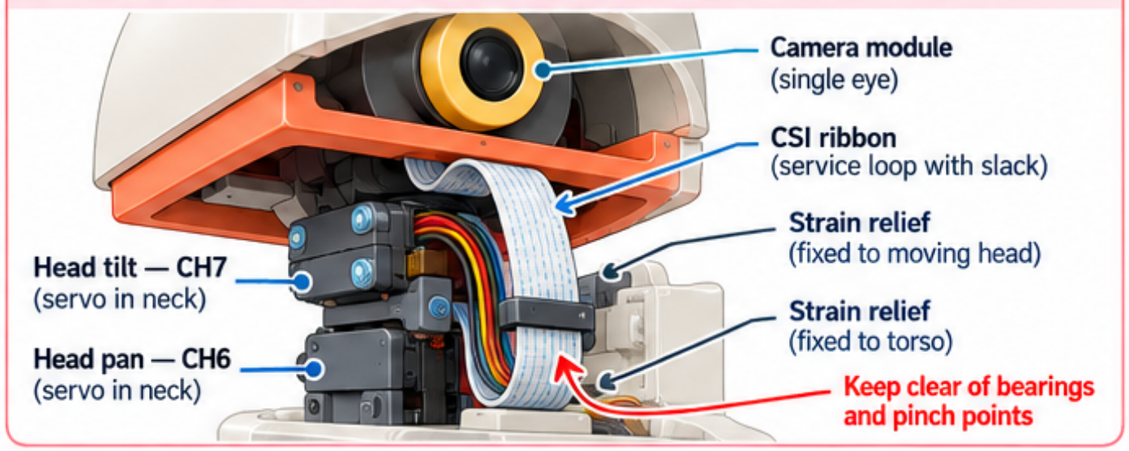
Proposed placement zones • 8 servos • mounts and clearances not verified



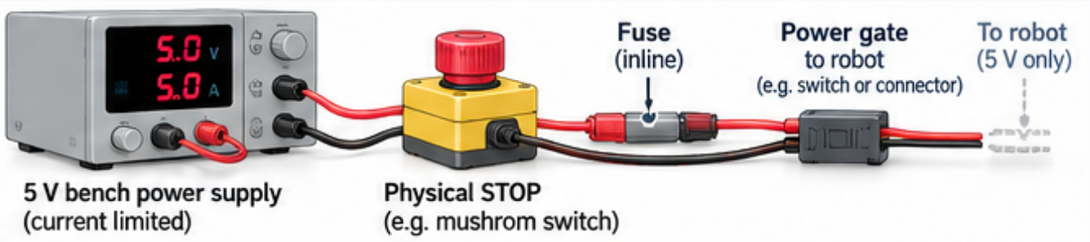
TORSO TRAY — COMPONENT PLACEMENT (ENLARGED VIEW)



NECK DETAIL — CAMERA CABLE AND SERVO PLACEMENT



EXTERNAL POWER AND STOP — KEEP REACHABLE (BENCH SETUP)



Placement concept only. Verify actual board sizes, servo torque, mounting coupons and cable travel before making parts. Keep servo power disconnected.

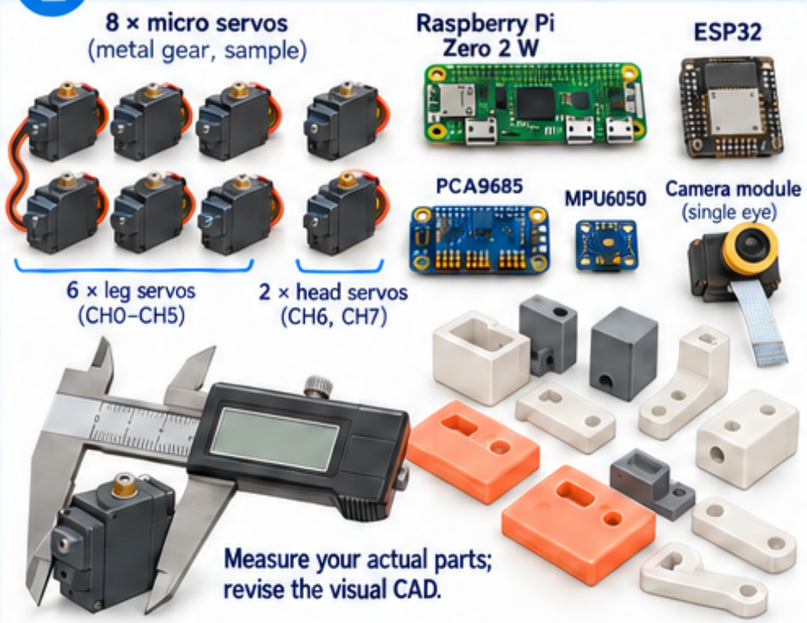
= structure (ivory) = feet / jaw (orange) = logic board (teal) = data (blue) = servo power (red)

MicroDuck visual reference: Pollen Robotics • CC BY-NC-SA 4.0

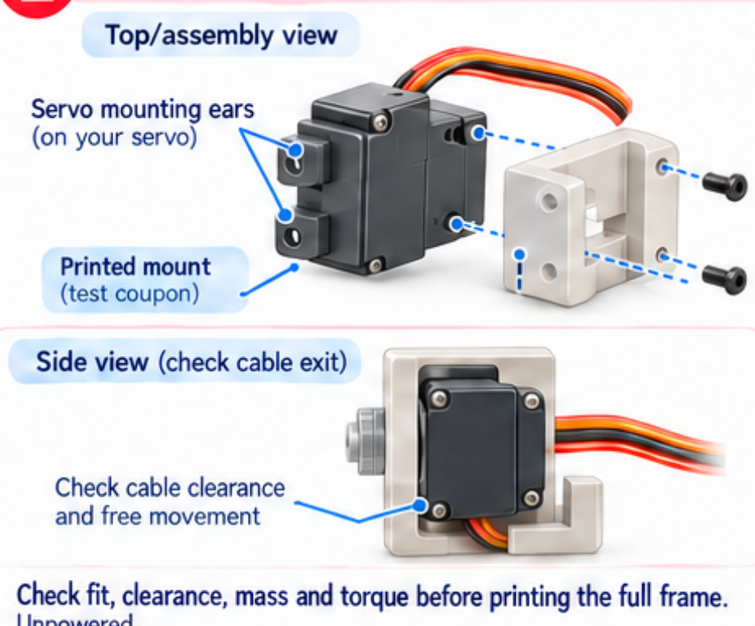
MICRODUCK DIY | SHEET 2 – MECHANICAL DRY-FIT

Steps 1–6 • unpowered fit checks • no verified mounting dimensions

1 INVENTORY + MEASURE

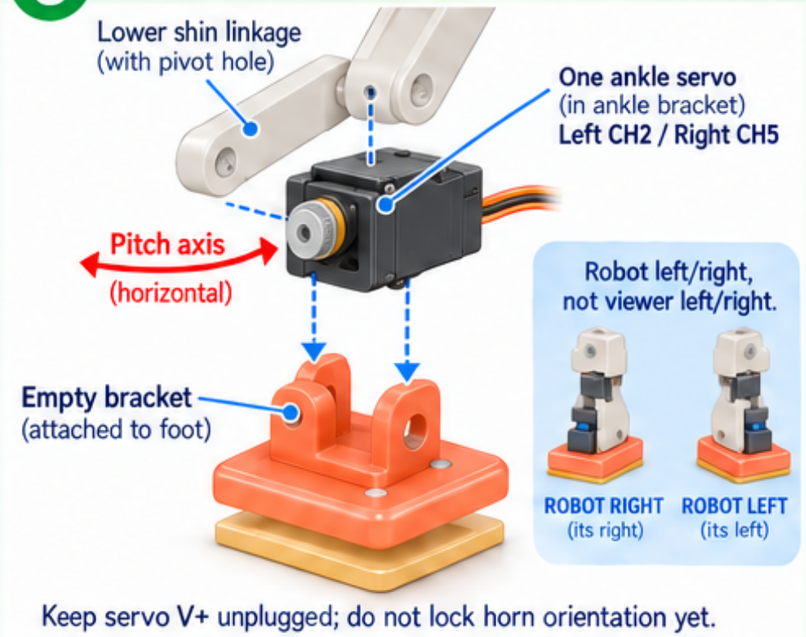


2 TEST ONE MOUNT



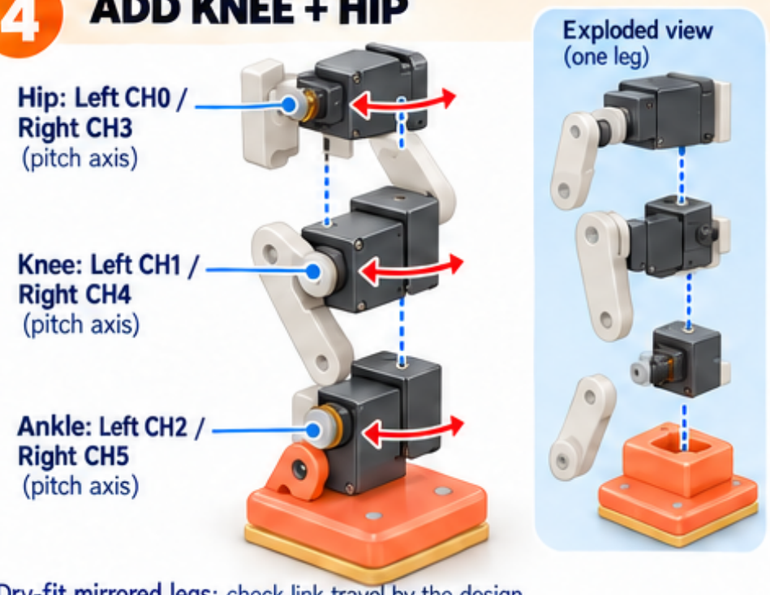
Check fit, clearance, mass and torque before printing the full frame.
Unpowered.

3 FOOT + SHIN DRY-FIT



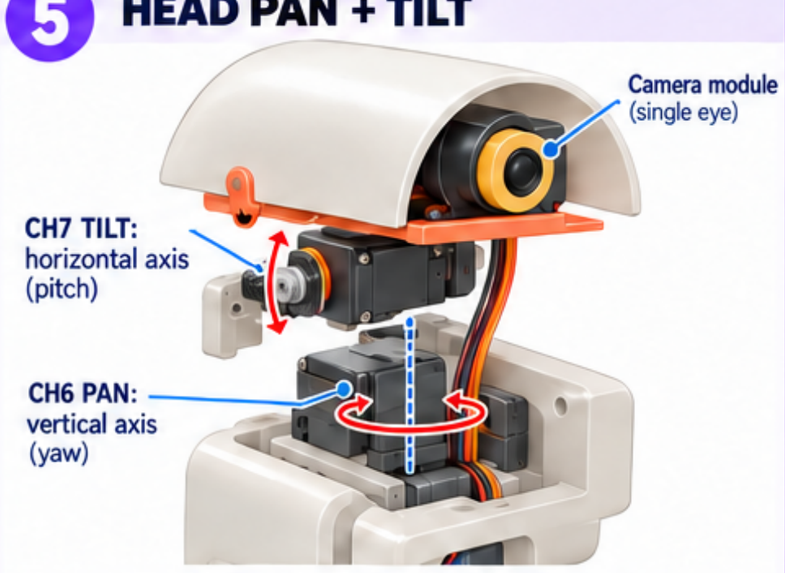
Keep servo V+ unplugged; do not lock horn orientation yet.

4 ADD KNEE + HIP



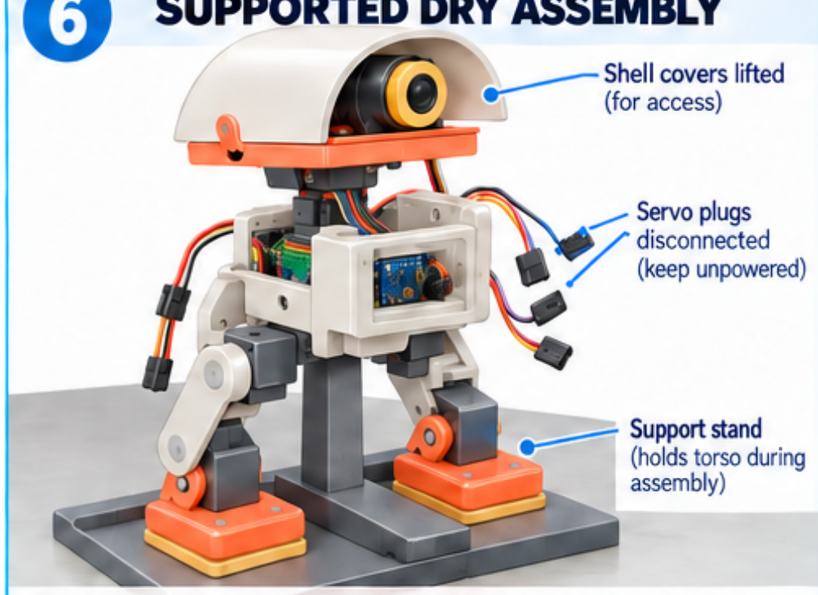
Dry-fit mirrored legs; check link travel by the design.
Three pitch axes, all parallel (horizontal). No hip-yaw or hip-roll actuators.

5 HEAD PAN + TILT



Test brackets and clearance; keep Pi in torso. Only **two** head servos (CH6 pan, CH7 tilt).

6 SUPPORTED DRY ASSEMBLY



Check mass, interference and cable paths.
Horns fit only after later measured calibration.

CURRENT BUILD STOPS AT DRY-FIT. Powered servo calibration requires the future commissioning/PWM module and verified power interlocks. This is not a validated printable assembly.

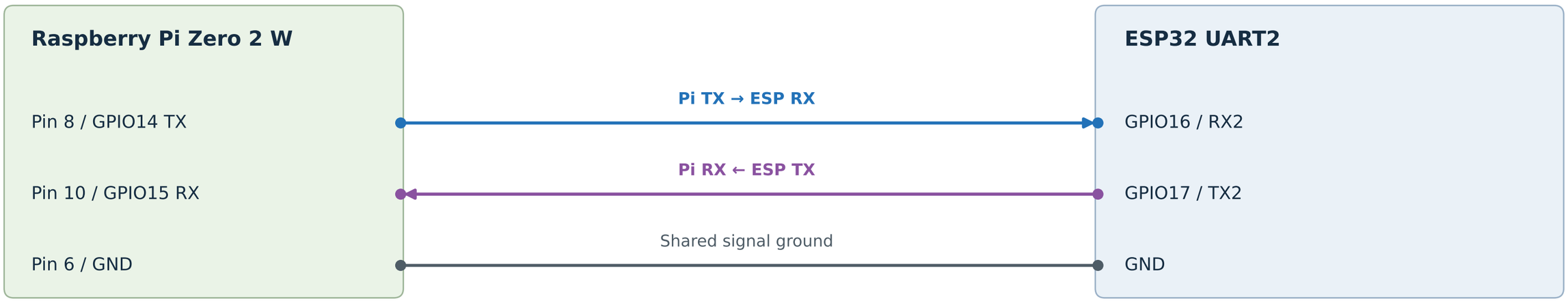
MicroDuck visual reference: Pollen Robotics • CC BY-NC-SA 4.0

03 / LOGIC WIRING — EXACT CONNECTIONS

Classic ESP32-WROOM DevKit only. Leave the servo V+ supply physically disconnected for this sheet.

A. Pi ↔ ESP32: three signal connections

115200 baud • 8N1 • 3.3 V signals



Power Pi + ESP32 through their approved USB/5 V logic inputs. Disable Pi serial login before using the UART.

B. ESP32 I²C: connect each row as one shared electrical net

SIGNAL	ESP32 / SOURCE	PCA9685 — 0x40	MPU6050 — 0x68
SDA / data	GPIO21	SDA	SDA
SCL / clock	GPIO22	SCL	SCL
LOGIC POWER	3.3 V logic source*	VCC = 3.3 V	3.3 V-compatible input*
REFERENCE	GND	GND	GND

* Verify the actual breakout's supply labels/regulator requirements and the 3.3 V source capacity before wiring.

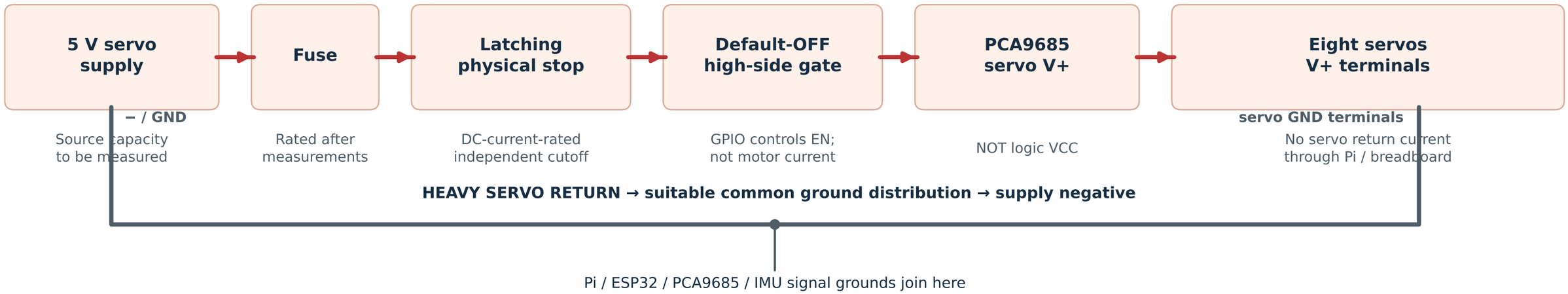
Keep I²C pull-ups at 3.3 V. AD0 low gives MPU6050 address 0x68. PCA9685 VCC is NOT servo V+.

No Pi SDA/SCL connection is used here. Match printed signal names on your own boards, not these drawing positions.

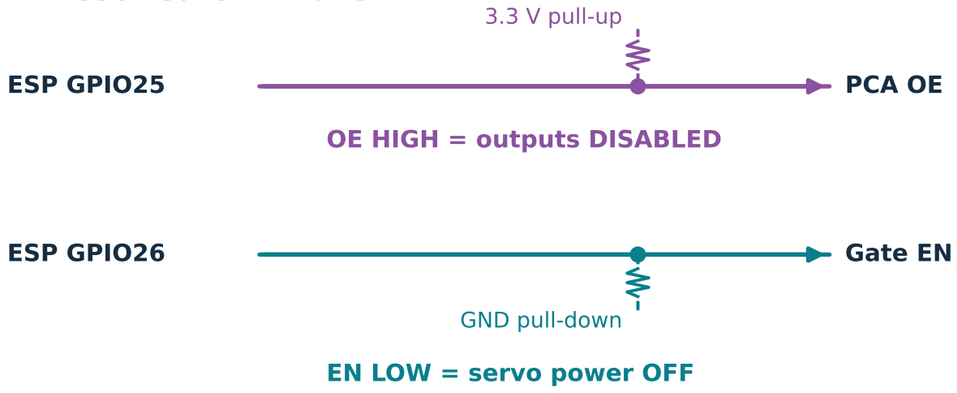
04 / SERVO POWER, INHIBITS & CHANNEL MAP

Planning diagram only. The current firmware has no PWM backend; never energize servos to try the software demo.

A. Dedicated servo positive supply — separate from the Pi/ESP32 positive rail



B. Reset-safe inhibits



External resistors required; values/circuit depend on selected boards.

C. PCA9685 signal channel → joint

CH 0	left_hip_pitch	LEG
CH 1	left_knee	LEG
CH 2	left_ankle_pitch	LEG
CH 3	right_hip_pitch	LEG
CH 4	right_knee	LEG
CH 5	right_ankle_pitch	LEG
CH 6	head_pan	HEAD
CH 7	head_tilt	HEAD

Left/right means the robot's own left/right, not the viewer's.

D. Identify the three actual servo connections before plugging in

SIGNAL → assigned CH **V+ → switched servo 5 V** **GND → heavy return**

Header order and servo lead colors vary: use board labels and the servo datasheet. Do not infer pin order from this sheet.

Before any future output: measured calibration, rated power/stop/gate, supported mechanism and commissioning checks are required.